SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Subject: Programming with Java	**Sem: III AY: 2026-27**
Assignment No: 14	
TITLE :Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.	

Problem Statement:

Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives

To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory

Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

Exercises:

1.Create an ATM program handling invalid withdrawal amount using try-catch.

Users > sumit > Documents > java > java1 > ATM.java > ATM

```java
import java.io.*;

public class ATM
{
    Run | Debug
    public static void main(String[] args)
    {
        try
        {
            int balance = 5000;
            int withdrawal = 6000;

            if(withdrawal > balance)
            {
                throw new Exception("Invalid withdrawal amount");
            }

            System.out.println("Withdrawal successful.");
            System.out.println("Remaining Balance: " + (balance - withdrawal));
        }
        catch(Exception e)
        {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Output:

```
[Running] cd "/Users/sumit/Documents/java/java1/" && javac ATM.java && java ATM
Error: Invalid withdrawal amount

[Done] exited with code=0 in 0.303 seconds
```

2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

Users > sumit > Documents > java > java1 > OnlineShopping.java > OnlineShopping > main(String[])

```java
import java.io.*;

public class OnlineShopping
{
    Run | Debug
    public static void main(String[] args)
    {
        try
        {
            int quantity = 0;

            if(quantity <= 0)
            {
                throw new Exception("Invalid product quantity");
            }

            System.out.println("Product quantity: " + quantity);
            System.out.println("Order placed successfully.");
        }
        catch(Exception e)
        {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

Output:

```
[Running] cd "/Users/sumit/Documents/java/java1/" && javac OnlineShopping.java && java OnlineShopping
Error: Invalid product quantity

[Done] exited with code=0 in 0.299 seconds
```

Conclusion:

The programs successfully demonstrate exception handling using try and catch blocks in Java. The ATM program handles an invalid withdrawal amount, while the Online Shopping program handles a product quantity less than or equal to zero. The catch block displays appropriate error messages when exceptions occur. These programs show how exception handling prevents abnormal termination and makes applications more reliable.